

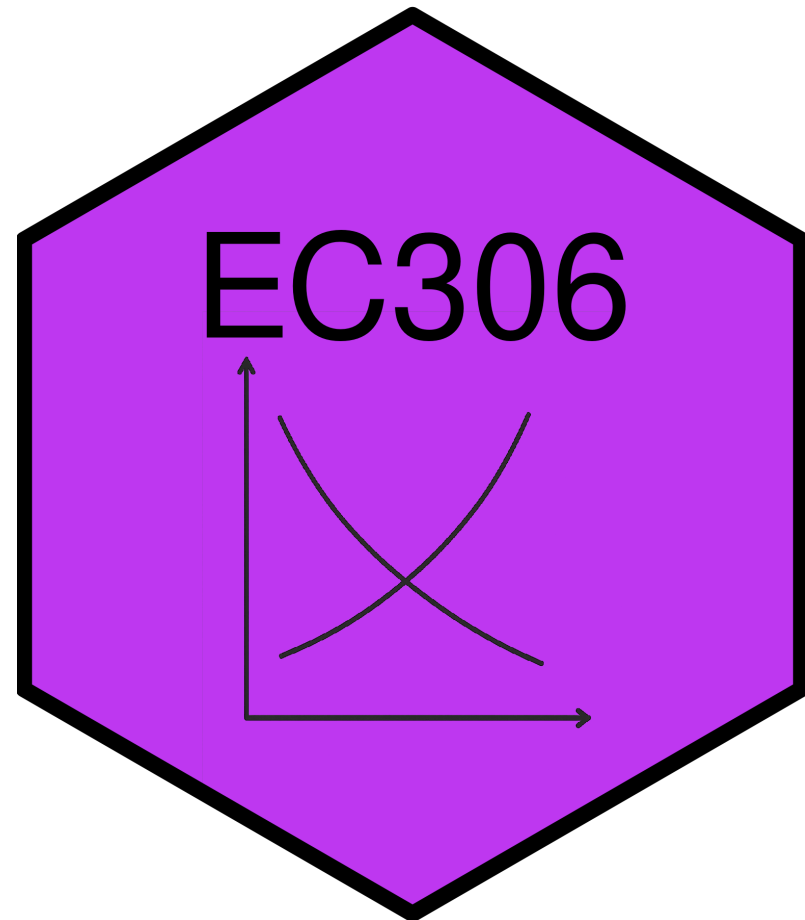
Labour Supply and Income Support Programs

EC306- Wages and Employment

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Goals of This Section

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In this section we will:

- Describe the main income support programs in Canada
- Show the effect of these programs on work decisions using the labour supply model
 - Demogrants, social assistance, negative income tax, wage subsidies, refundable tax credits
- Examine statistical evidence for the model predictions

Income Maintenance Programs

Overview of Income Maintenance Programs

- Represent about 10% of GDP in Canada
- Primary aims:
 - Raise income of specific groups (parents, elderly)
 - Supplement low wages
- Major programs: child benefits, employment insurance, social assistance, pensions

! Key Challenge

Balancing income support with work incentives

Government Transfers in Canada

In 2017, 83% of families received some form of government transfer:

Program	% Receiving	Average Amount
Federal Child Benefits	20.2%	\$6,314
Employment Insurance	14.0%	\$8,352
Social Assistance	11.3%	\$8,367
Old Age Security	25.1%	\$8,566
CPP/QPP	32.4%	\$10,092

Issues in Program Design

Universal vs. Targeted Programs

- Universal: easy to implement but expensive
- Targeted: benefits only those in need but costly to administer

Work Disincentives

- Some programs reduce incentive to work
- Example: welfare benefits can encourage labour force dropout for those with strong preference for leisure

Permanent vs. Transitory Income Loss

- Different situations require different program designs
- Temporary layoff → short-term support
- Long-term disability → longer support + reintegration costs

Think About It

If 83% of Canadian families receive some form of government transfer, why is program design so important? What happens if a program is too generous or too stingy?

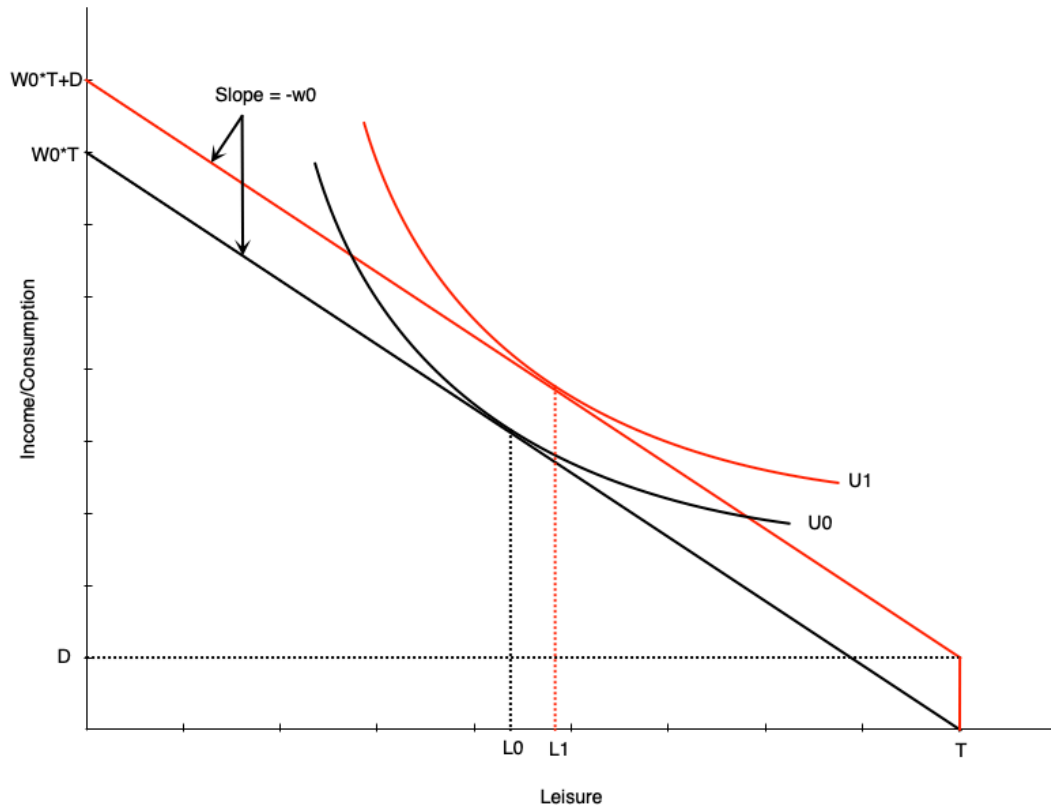
Demogrants, Welfare, Wage Subsidies

Four Standard Income Support Programs

We will examine four policy approaches:

1. [Demogrants](#) - Fixed payment to all regardless of income/employment
2. [Welfare](#) - Lump-sum payment for non-workers, reduced dollar-for-dollar with earnings
3. [Negative Income Tax \(NIT\)](#) - Welfare payment that decreases at rate $< 100\%$ with earnings
4. [Wage Subsidies](#) - Addition to the wage rate

Demogrant: Theory



How it works:

- Unconditional lump sum transfer D
- Shifts budget line up parallel
- Wage unchanged → **pure income effect**

Labor supply impact:

- If leisure is normal → optimal leisure rises
- Reduces hours worked
- May induce non-participation

⚠ Work Incentive Effect

Unambiguously negative

Demogrant: Canadian Example

Universal Child Care Benefit (UCCB)

- Paid to families with children under 18
 - \$160/month per child under 6
 - \$60/month per child aged 6-17
- Replaced in 2016 by Canada Child Benefit (CCB)
 - More targeted to low/middle income families

Evidence (Schirle, 2015)

UCCB reduced labour supply of women with children under 6 by ~3.2 percentage points for low-educated married women

Welfare: Overview

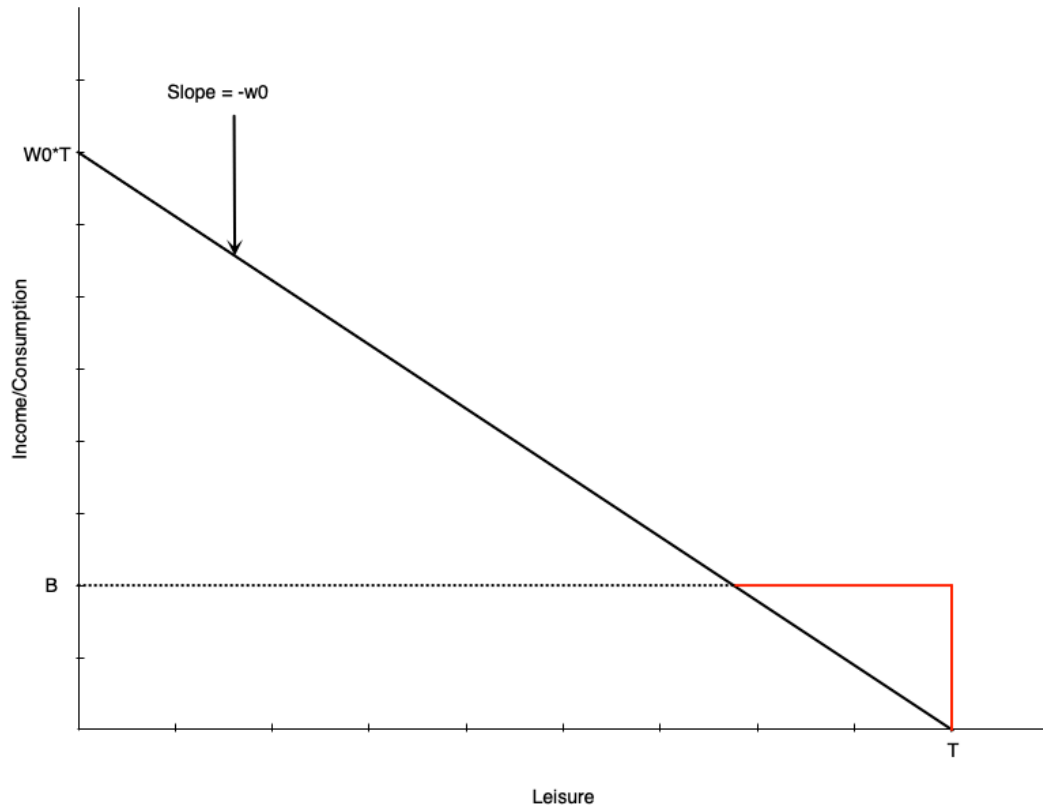
Welfare = Social/Income Assistance in Canada

- Lump-sum payment to non-workers
- Reduced **dollar-for-dollar** with earned income

! Historical Criticism

Creates negative work incentives for some, but must balance against need for income support

Welfare: Budget Constraint



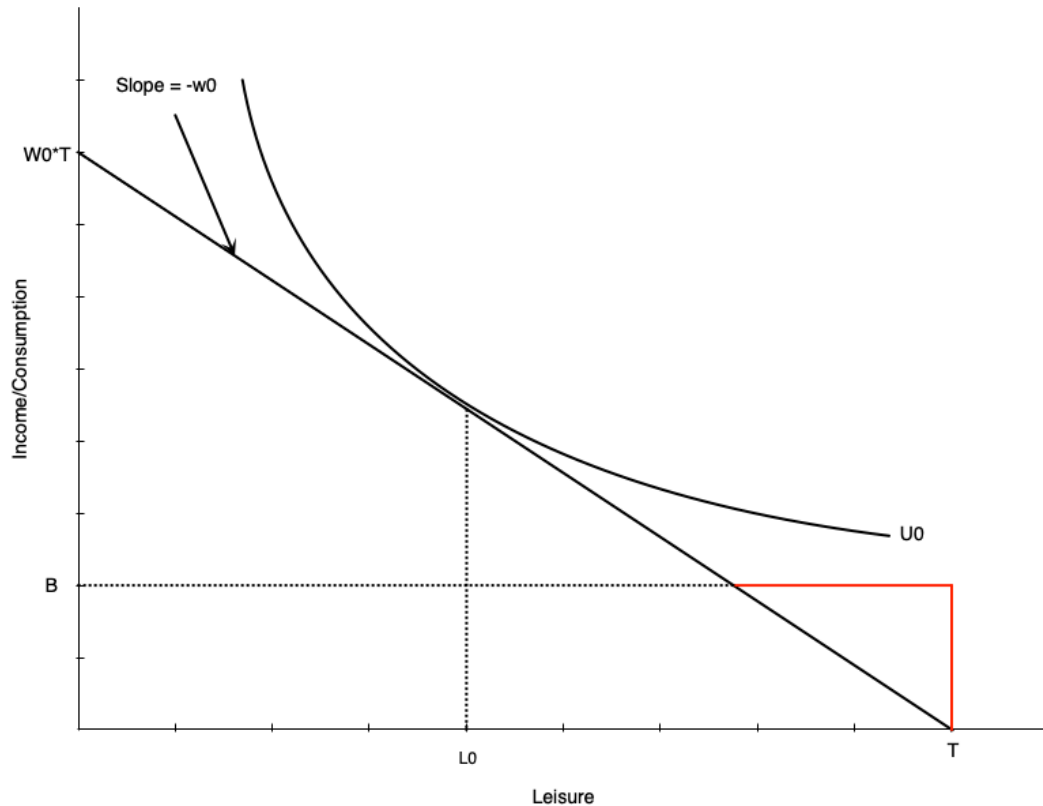
Budget line structure:

- Black line: original budget (leisure at L_0)
- Red then black: with welfare benefit B

Key features:

- Benefit B at zero hours of work
- Reduced \$1 for each \$1 earned
 - Equivalent to 100% tax rate on earnings
- Income constant at B until earned income equals benefit
- After that, earn wage w_0 per hour worked

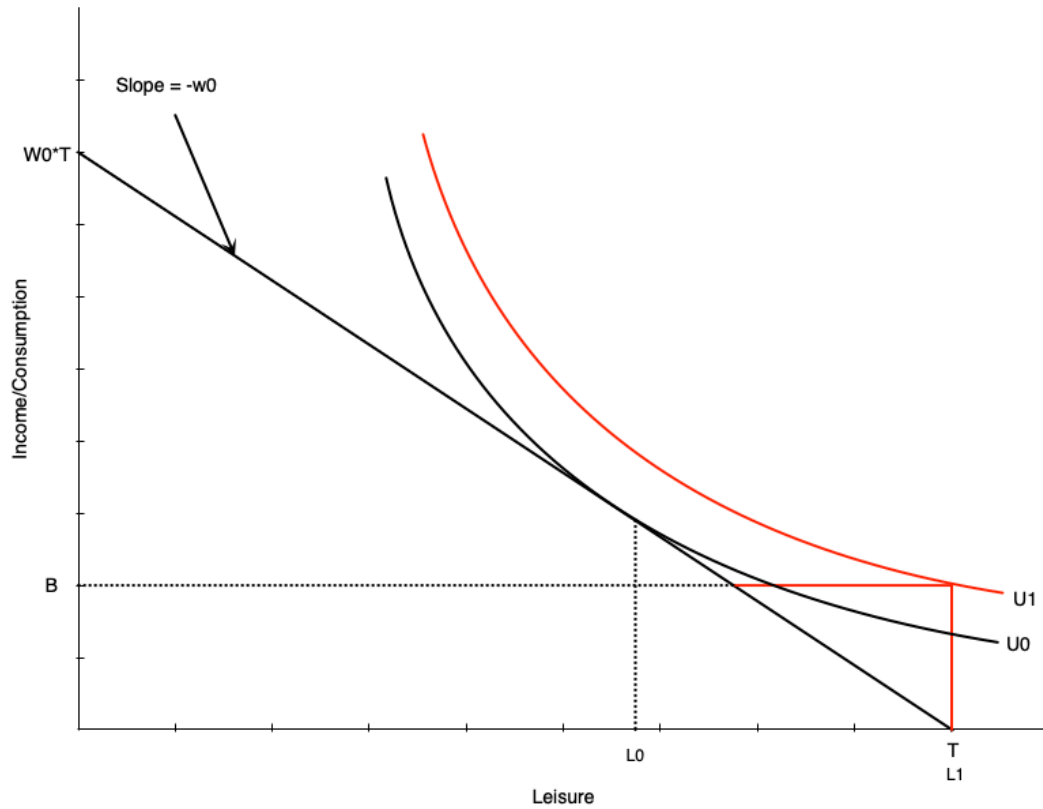
Welfare: No Effect Case



For some workers, welfare has no effect:

- Person working H_0 hours before benefit
- Still maximizes utility at H_0 hours after
- Generally occurs with people already working many hours

Welfare: Work Disincentive Case



Others face strong work disincentive:

- Person chooses **zero hours** after welfare benefit

This occurs when:

- Welfare benefit is generous
- Individual places high value on leisure

Warning

This is the problematic effect of welfare programs

Discussion

Why might someone who was working 20 hours per week before welfare was introduced choose to stop working entirely, while someone working 40 hours per week continues to work the same amount?

Combating Work Disincentives

Policy options:

1. Reduce benefit level

- But may not provide adequate support

2. Increase wages

- Through minimum wages, subsidies, training, unions
- Can be expensive and may reduce labour demand

3. Change preferences

- Alter attitudes toward work/leisure
- Difficult to achieve

4. Negative income tax

- Reduce tax-back rate on earnings
- Discussed next...

Welfare: Evidence from Quebec

Natural experiment: Welfare benefit structure changed in 1989

Before 1989:

- Age-based benefits for singles/couples with no kids
- Under 30: \$185/month
- 30 and over: \$507/month

Results at age 30:

- Much longer time on welfare
- Lower employment rates

Note

Turning 30 created a **large discontinuous jump** in benefits

Negative Income Tax: Overview

What is NIT?

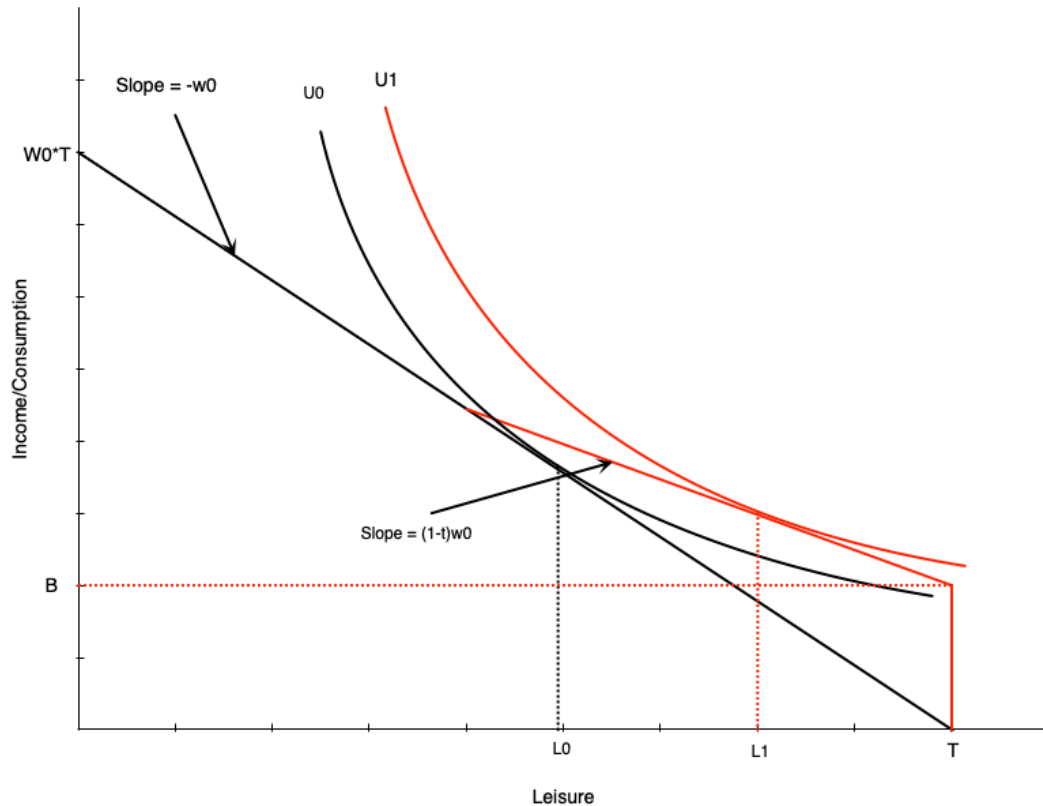
- Welfare program where benefit reduced at rate $< 100\%$ of earnings
- Example: \$10,000 benefit reduced by 50¢ per dollar earned

Advantages

Provides income support with fewer work disincentives. Still maintains incentive to work since benefit reduces slowly.

Also known as: Guaranteed Annual Income (GAI) or Basic Income

Negative Income Tax: Budget Constraint



NIT has two components:

- Benefit B at zero hours of work
- Tax rate t on earned income (until earned income equals benefit)

Effect on labour supply:

- Pivots budget line up vs. pure welfare
- Work hours reduced from H_0 to H_1
- Person still works some hours

❗ Important

Work disincentive less than pure welfare

Negative Income Tax: Income and Substitution Effects

Income effects:

- Benefit increases non-labour income $\rightarrow$ increases leisure $\uparrow$
- Tax has negative income effect $\rightarrow$ reduces leisure $\downarrow$
- **Net:** benefit effect dominates $\rightarrow$ leisure $\uparrow$

Substitution effect:

- Tax rate reduces effective wage
- Makes leisure relatively cheaper $\rightarrow$ reduces work hours $\downarrow$

Overall Effect

Negative effect on work hours, but **smaller than pure welfare**

Negative Income Tax: Math

The budget constraint with NIT:

$$C = \begin{cases} w_h + B - tw_h, & \text{if } B - tw_h \geq 0 \\ w_h, & \text{otherwise} \end{cases}$$

Interpretation:

- Consumption = earned income (w_h) + net benefit ($B - tw_h$) if net benefit > 0
- Otherwise just earned income (w_h)

Break-even point: benefit reaches zero when $w_h = \frac{B}{t}$

MINCOME: Manitoba's NIT Experiment

Program details (1970s):

- Randomized controlled trial
 - Treatment group received NIT
 - Control group did not
- Conducted in Dauphin (saturated - everyone got NIT) and Winnipeg
- 7 treatment groups with different benefit levels and tax rates
- Ended in 1979 due to political changes

MINCOME Treatment Groups

		Tax Rate (on total Income)		
		35%	50%	75%
Guarantee at enrolment	\$3800	Plan 1	Plan 3	*
	\$4800	Plan 2	Plan 4	Plan 7
	\$5480	X	Plan 5	Plan 8
		Plan 9		
* Plan 6 was collapsed into Plan 7 because participants viewed this option as too punitive and left the experiment. X This option was deemed to expensive for the budget. Plan 9 was the control group				

Source: Wikipedia

MINCOME: Results

Labour supply effects:

- Small decline in hours worked
 - Mainly mothers of newborns
 - Women in two-parent families
 - No significant decline for men

Health outcomes:

- Fewer hospitalizations and doctor visits
- Especially for mental health and accidents

Education:

- Increase in adolescents finishing high school

Wage Subsidies: Theory

What is a wage subsidy?

- Payment to workers that increases their wage rate
- Example: \$5/hour subsidy increases \$15/hour wage to \$20/hour

Effect on budget constraint:

- Budget line pivots upward

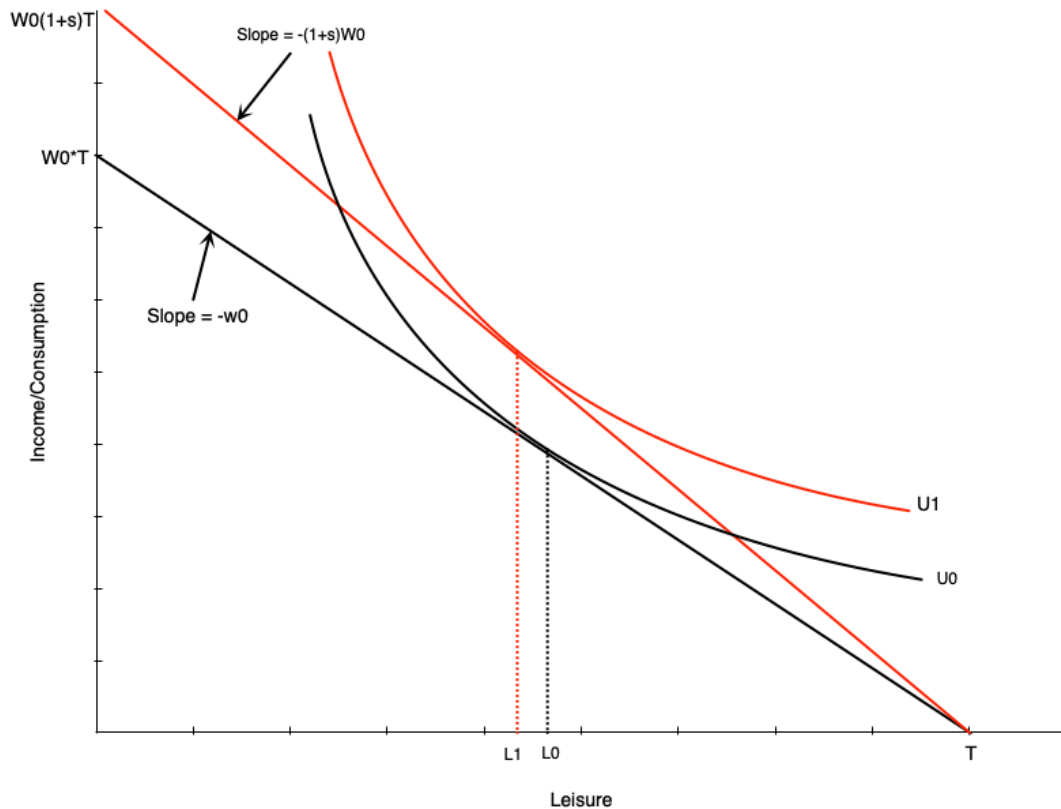
Labour supply effects:

- Income effect → increases leisure ↑
- Substitution effect → decreases leisure ↓

Warning

Overall effect on work hours is **ambiguous**

Wage Subsidies: Diagram



Budget line pivots up with subsidy

- Graph shows person working more
- But effect is theoretically ambiguous
- Income and substitution effects work in opposite directions

Wage Subsidies: Policy Implications

Advantages

- Less negative work incentives than welfare or NIT
- No reduction in effective wage rate

Disadvantages

- Do not provide income support to non-workers
- May not help those in most need

In Canada:

- Pure wage subsidies mostly go to **employers**
- Unclear if any benefits passed on to workers

Think About It

A negative income tax reduces the benefit as you earn more, while a wage subsidy increases your pay for each hour worked. Which program would you expect to have a larger positive effect on labour force participation among non-workers, and why?

Tax Credits: Overview

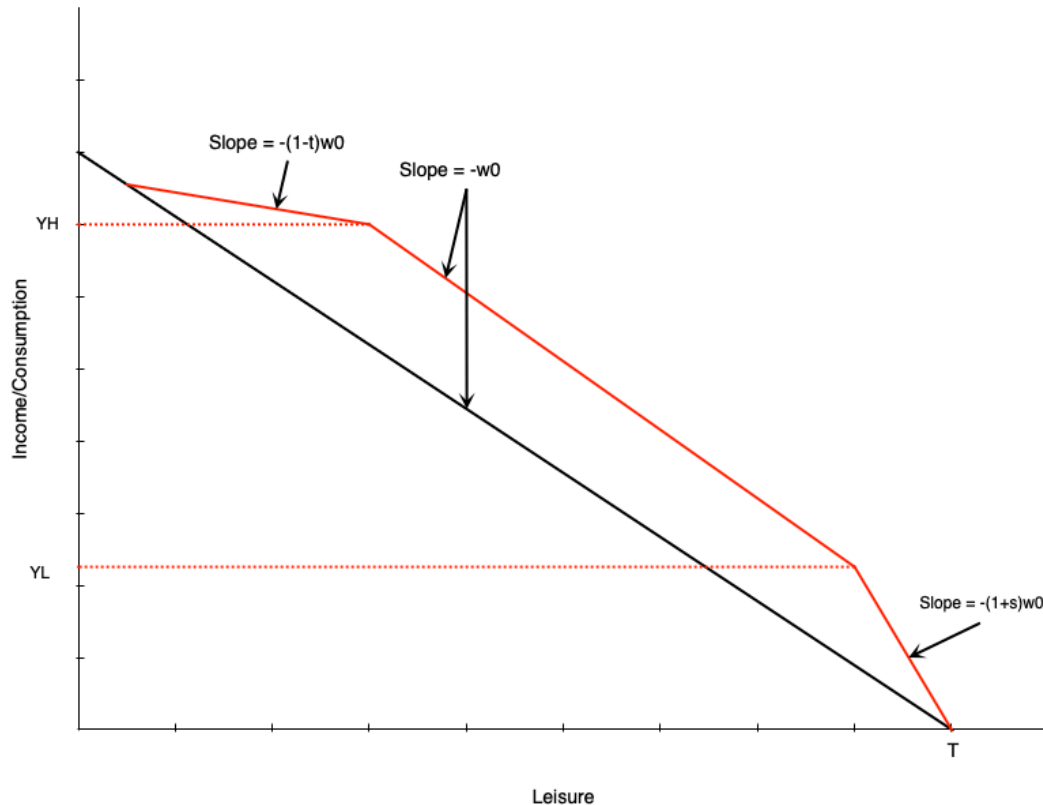
Two types of tax credits:

1. Non-refundable: Can reduce tax owed to zero (but not below)
2. Refundable: Can reduce tax owed below zero (results in payment)

Income Maintenance Programs via Refundable Tax Credits

- [Canada Worker Benefit \(CWB\)](#) - Canada
- [Earned Income Tax Credit \(EITC\)](#) - US

Tax Credits: Budget Constraint



Three segments:

1. **Phase-in**: Subsidy proportional to earned income
2. **Plateau**: Maximum benefit level
3. **Phase-out**: Benefit reduced with earned income

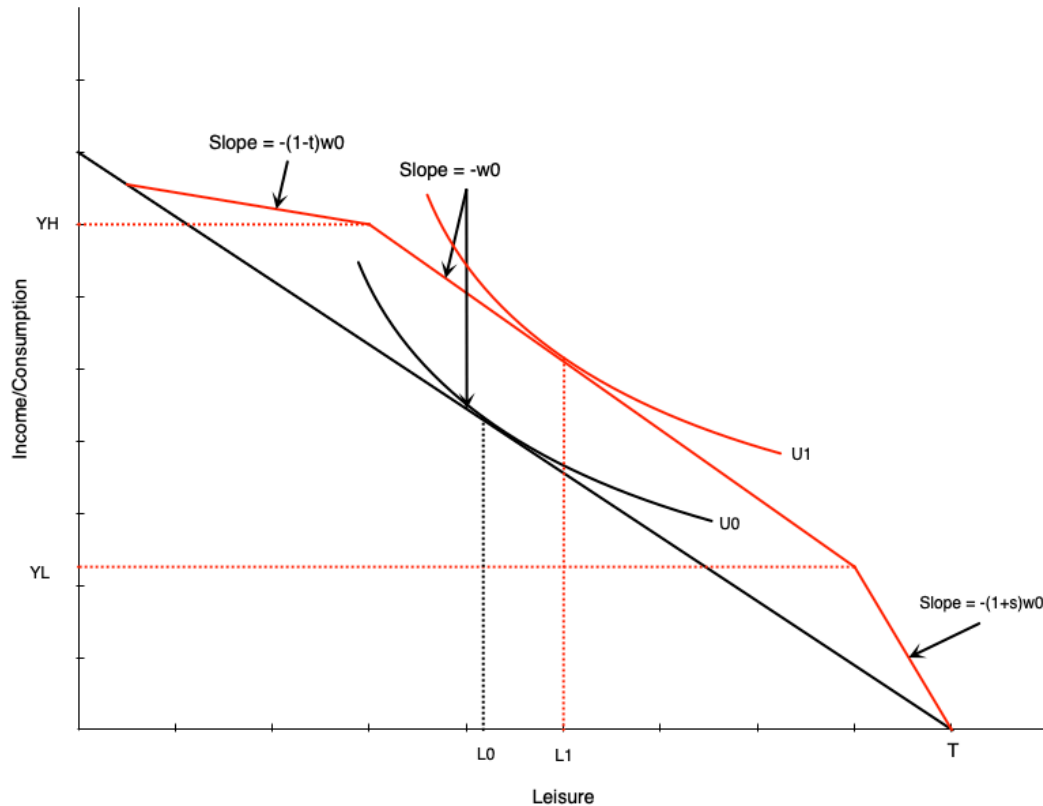
Combines elements of:

- Wage subsidy (phase-in)
- Demogrant (plateau)
- NIT (phase-out)

Tax Credits: Effect in Plateau Range

Person initially in middle range:

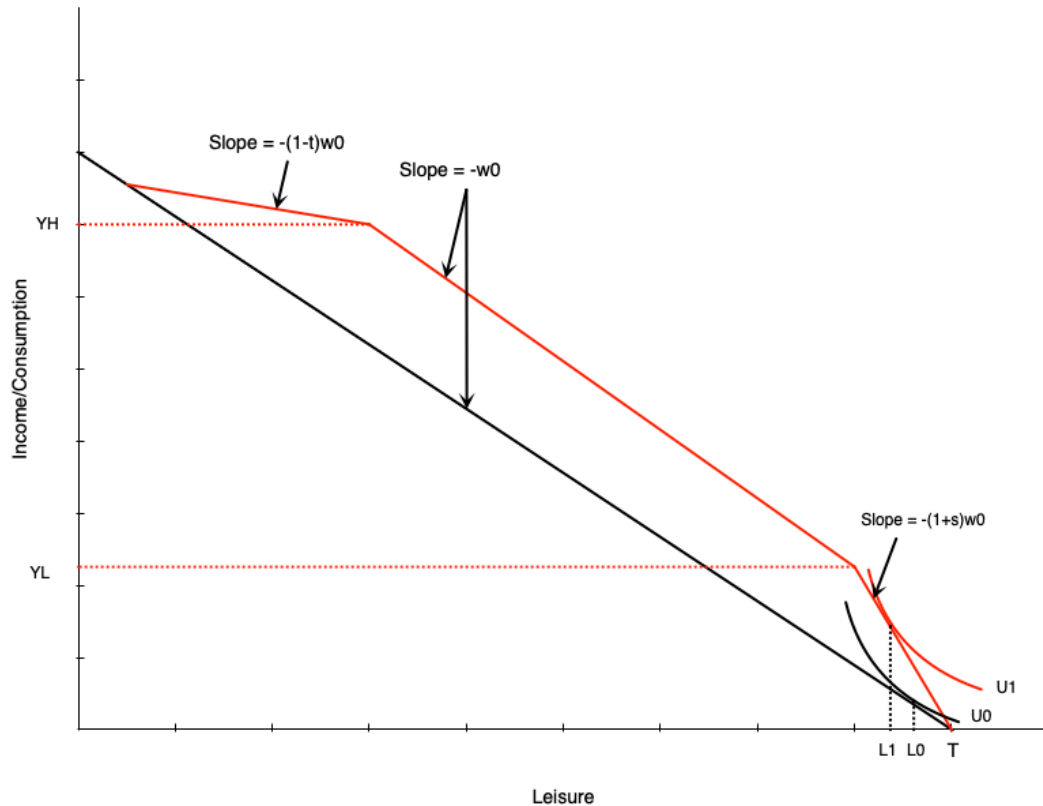
- Tax credit gives maximum benefit
- Acts like **pure income effect**
- If leisure is normal good $\rightarrow$ reduce work hours



Tax Credits: Effect in Phase-In Range

Person in phase-in range:

- Sees increase in effective wage
- Acts like **pure wage subsidy**
- Income and substitution effects in opposite directions



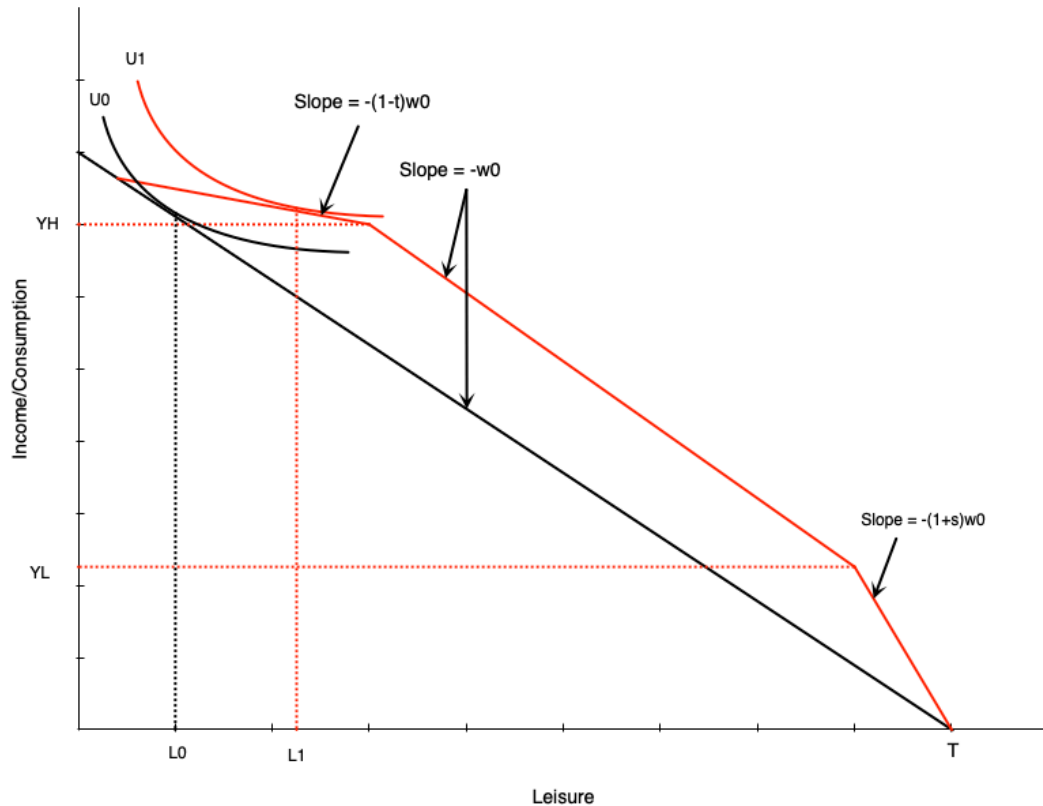
Warning

Effect on hours is **ambiguous**

Tax Credits: Effect in Phase-Out Range

Person in phase-out range:

- Similar to **negative income tax**
- Income effect $\rightarrow$ reduce hours $\downarrow$
- Substitution effect $\rightarrow$ reduce hours $\downarrow$



❗ Important

Effect on work is **unambiguously negative**

TODO — The Self-Sufficiency Project

 Not yet written

BFGLSS 2026 covers:

- A large Canadian earnings-supplement experiment, listed in Chapter 4 under *Empirical Evidence of Incentive Effects*. No slides exist for it.

Summary

Summary

Income support programs:

- Provide crucial income support to various groups
- Can create work disincentives in some cases
- Must balance support needs against incentive effects

We examined several programs through the labour supply model:

- Demogrants, welfare, negative income tax
- Wage subsidies and refundable tax credits